

pre-test questions the basis of questions to be presented per topic in term 3:

Group 1· PROJECTILE MOTION (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. Any object thrown, kicked, or launched into the air and acted on only by gravity and air resistance is called a:

- * A. Linear object
- * B. Stationary body
- * C. Circular object
- * *D. Projectile* ☒

2. The curved path a projectile follows through the air is its:

- * *A. Trajectory* ☒
- * B. Radius
- * C. Slope
- * D. Axis

3. The two independent motions that combine to make projectile motion are:

- * A. Up and down only
- * *B. Constant horizontal motion + accelerated vertical free-fall motion* ☒
- * C. Left and right only
- * D. Fast and slow motion

4. If you kick a football with the same force, which angle will make it go the farthest horizontal distance (maximum range) on flat ground?

- * A. 75°
- * *B. 45°* ☒
- * C. 15°
- * D. 30°

5. If you want a basketball shot to go as high as possible into the air, you should release it at an angle closest to:

- * A. 45°
- * *B. 90° (straight up)* ☒
- * C. 30°
- * D. 0° (horizontal)

6. If you throw a ball horizontally and drop another identical ball from the same height at the same time, which hits the ground first?

- * A. Thrown ball hits first
- * B. Dropped ball hits first
- * *C. Both hit at exactly the same time* ☒
- * D. Depends on color

7. The horizontal distance a projectile travels from launch to landing is its:

- * A. Trajectory
- * *B. Range* ☒
- * C. Height
- * D. Velocity

8. The maximum vertical distance the projectile reaches above launch level is its:

- * *A. Maximum height* ☒
- * B. Trajectory
- * C. Range
- * D. Angle

9. If you shoot two basketballs at the same angle, the one thrown with more speed will have:

- * A. Same range and height
- * B. Less height only
- * C. Shorter range only
- * *D. Both longer range and greater height* ☒

10. When air resistance is ignored, the only force acting on a projectile after release is:

- * A. Wind forward
- * B. Magnetism
- * *C. Gravity (pulling downward)* ☒
- * D. Your hand

11. A sepak takraw player kicks the ball at 30° , another kicks at the same speed at 60° . Their ranges will be:

- * A. 60° goes twice as far
- * B. 30° goes farther
- * C. Both fall at the same spot as 45°
- * *D. Almost exactly equal (complementary angles)* ☒

12. Between the same two complementary kicks above, which one reaches a higher maximum height?

- * A. Same height
- * *B. 60° kick* ☒
- * C. Neither rises
- * D. 30° kick

13. Why does a basketball player use a high arc (large angle) when shooting from far away?

- * *A. To give the ball more time in the air to reach the hoop and enter the ring from above* ☒
- * B. To use less force
- * C. To avoid gravity
- * D. To make it go faster

14. If you kick a football at 20° and another at 70° with the same speed, which has the shortest time of flight?

- * A. Same time
- * *B. 20° kick* ☒
- * C. 70° kick
- * D. Both stay forever

15. A projectile's horizontal velocity component (ignoring air resistance) during flight:

- * A. Becomes zero at the top
- * B. Decreases due to gravity
- * *C. Stays exactly constant* ☒
- * D. Increases due to gravity

16. A projectile's vertical velocity component at the exact top of its arc is:

- * A. Maximum upward
- * B. Equal to launch speed
- * C. Maximum downward
- * *D. Zero (momentarily stops going up)* ☒

17. In real life (with air resistance), a real football's actual range compared to the ideal physics prediction will be:

- * *A. Slightly shorter* ☒
- * B. Exactly the same
- * C. Longer
- * D. Twice as far

18. A long jumper wants maximum range. What combination should they aim for?

- * A. Very low angle only
- * *B. Fast run-up + launch angle near 45° * ☒
- * C. Jump straight up
- * D. Very high angle only

19. If you double the launch speed of a projectile at 45° , its range becomes approximately:

- * A. Same
- * B. $2\times$ longer
- * *C. $4\times$ longer* ☒
- * D. $8\times$ longer

20. Two identical stones are thrown: A at 10 m/s, 45° ; B at 20 m/s, 45° . Which statement is true?

- * A. Same flight time
- * *B. B goes $4\times$ farther, about $4\times$ higher, and about $2\times$ longer in the air* ☒
- * C. Same range
- * D. Same maximum height

21. Which is the correct full relationship among the four projectile variables?

- * A. Mass of the object is the biggest factor
- * *B. Range and height depend on both launch speed and launch angle through horizontal and vertical velocity components* ☒
- * C. Only speed matters
- * D. Only angle matters

22. At what point in a projectile's full flight is its total speed slowest?

- * A. Same speed everywhere
- * *B. At the very peak (top of the trajectory)* ☒
- * C. Right after launch
- * D. Right before landing

23. A basketball is released and lands at the same height. At what two points in its path is its speed exactly equal?

- * A. Launch vs. top
- * B. Never equal
- * *C. Any two points at the same height (e.g., halfway up and halfway down)* ☒
- * D. Top vs. landing

24. Why do professional archers aim slightly above the target even at short distances?

- * A. To make it go faster
- * B. Wind always blows up
- * *C. Gravity pulls the arrow downward during flight, so they must compensate for the drop* ☒
- * D. Tradition

25. If you launch from ground level, which pair of angles will give the same range but very different heights and flight times?

- * A. 10° and 20°
- * B. 40° and 50°
- * C. 25° and 65°
- * *D. Both B and C* ☒

26. A boy kicks a ball horizontally off a 10 m cliff while another drops an identical ball from the same height at the same instant. Which hits first?

- * A. Kicked ball first
- * *B. Both hit the ground at exactly the same time* ☒

- * C. Kicked ball never lands
- * D. Dropped ball first

27. For a fixed launch speed, as angle increases from $0^\circ \rightarrow 45^\circ \rightarrow 90^\circ$, how do range and height change?

- * A. Both always decrease
- * B. Range constant; height rises
- * *C. Range rises to a maximum at 45° then falls; height keeps increasing to 90° * ☒
- * D. Both always increase

28. The only reason 45° gives maximum range on flat ground is:

- * A. Air resistance is zero only at 45°
- * B. Gravity is strongest there
- * *C. It perfectly balances time of flight and horizontal speed so their product is maximized* ☒
- * D. It is the middle number

29. In a projectile investigation using a marble launcher, a student keeps the angle fixed at 45° and increases compression (speed) each trial. They will observe:

- * A. Range stays the same
- * B. Height stays the same
- * C. Range increases linearly
- * *D. Range increases roughly as the square of the compression distance* ☒

30. Which everyday action is *NOT* an example of projectile motion?

- * *A. Driving a car in a straight flat road at constant speed* ☒
- * B. Throwing a frisbee (ignoring lift)
- * C. Kicking a penalty
- * D. Shooting a three-pointer

Group 2 · MOMENTUM AND COLLISIONS (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

Here's the same set with the *correct answers randomized* so there is *no obvious pattern* (not mostly *B*).

1. Momentum is defined mathematically as:

- * A. Mass $\times$ acceleration
- * *B. Mass $\times$ velocity ($p = m \times v$)* ☒
- * C. Force $\times$ distance
- * D. Weight $\times$ height

2. Momentum describes in physics what we mean in everyday life when we say:

- * A. An object is hot
- * B. An object is bright
- * *C. A moving object is hard to stop* ☒
- * D. An object is heavy

3. Which has more momentum: a slow truck or a fast bullet?

- * A. Always the truck
- * *B. Depends on BOTH mass AND velocity of each* ☒
- * C. Always the bullet
- * D. They always have equal momentum

4. A collision where objects bounce off perfectly with no kinetic energy lost is called:

- * A. Inelastic
- * B. Explosion

- * *C. Elastic* ☒
- * D. Plastic

5. A collision where objects stick together or deform, losing kinetic energy to heat and sound, is called:

- * A. Perfect rebound
- * *B. Inelastic* ☒
- * C. Conservative
- * D. Elastic

6. The Law of Conservation of Momentum says that if no outside force acts, total momentum before collision equals:

- * A. Zero always
- * B. Double before
- * C. Half before
- * *D. Total momentum after collision* ☒

7. To change an object's momentum, you must apply:

- * *A. A net force acting for some amount of time* ☒
- * B. Heat
- * C. Only mass
- * D. Only light

8. Seatbelts, airbags, and crumple zones all reduce injury mainly by:

- * A. Making the car lighter
- * B. Increasing force
- * *C. Increasing the stopping time, lowering the average force* ☒
- * D. Stopping instantly

9. If two identical jeepneys move at different speeds, the faster one has:

- * A. Same momentum
- * *B. More momentum* ☒
- * C. Zero momentum
- * D. Less momentum

10. When a moving billiard ball hits a stationary identical ball head-on and stops, the target ball:

- * A. Explodes
- * *B. Moves away with almost exactly the original ball's speed* ☒
- * C. Also stays still
- * D. Moves twice as fast

11. Which is the best everyday example of an almost elastic collision?

- * A. Two train cars coupling together
- * *B. Two billiard balls colliding on a table* ☒
- * C. Car hitting a wall
- * D. A ball of clay hitting the floor

12. Which is the best everyday example of a perfectly inelastic collision?

- * *A. Two train cars bump and lock together moving as one* ☒
- * B. Marbles colliding
- * C. Superball bouncing
- * D. Tennis ball on a racket

13. A 2000-kg jeepney and a 200-kg motorcycle both move at 10 m/s. Which is harder to stop?

- * A. Equally hard

- * B. Motorcycle
- * *C. Jeepney (10× more momentum)* ☒
- * D. Neither moves

14. A fast lightweight boxer's punch can still hurt a much heavier opponent because:

- * A. Gravity increases
- * B. Only mass matters
- * C. Mass doesn't matter
- * *D. High velocity gives the fist large momentum* ☒

15. Why is falling on a soft mattress less painful than falling on concrete from the same height?

- * A. Concrete is colder
- * B. The mattress reduces your mass
- * *C. The mattress increases stopping time, reducing force* ☒
- * D. Gravity is weaker on the mattress

16. When a cannon fires a cannonball forward, the cannon kicks backward. This is explained by:

- * A. Gravity
- * *B. Conservation of momentum* ☒
- * C. Elastic collision
- * D. Magic

17. In a car crash, crumple zones at the front help mainly by:

- * A. Looking nice
- * *B. Making the passenger cabin stop over a longer time* ☒
- * C. Reducing car mass
- * D. Making the car faster

18. Airbags work together with seatbelts because:

- * *A. Seatbelts hold you while airbags spread the force and increase stopping time* ☒
- * B. Airbags alone stop you instantly
- * C. They look good
- * D. Seatbelts are useless alone

19. Two identical balls move at equal speed toward each other, collide, and bounce back at the same speed. This collision is:

- * A. Explosive
- * *B. Very nearly elastic* ☒
- * C. Impossible
- * D. Perfectly inelastic

20. If you want to maximize the change in momentum of a kicked football, you should:

- * A. Tap it softly
- * B. Use a very light foot
- * C. Kick perpendicular to the motion
- * *D. Apply the largest possible force for the longest possible time* ☒

21. Which correctly explains why momentum depends on BOTH mass and velocity?

- * *A. A heavy slow object or a light fast object can both be equally hard to stop* ☒
- * B. Teacher said so
- * C. Mass cancels velocity
- * D. Only mass matters in collisions

22. In any collision—elastic or inelastic—which quantity is ALWAYS conserved?

- * A. Shape
- * B. Speed
- * *C. Total linear momentum of the isolated system* ☒
- * D. Kinetic energy

23. A 1-kg clay blob moving at 5 m/s hits and sticks to a 4-kg stationary block. Afterward the combined mass moves at:

- * A. 0 m/s
- * B. 5 m/s
- * *C. 1 m/s* ☒
- * D. 2.5 m/s

24. The main physics reason airbags reduce fatalities so dramatically is:

- * A. They block glass
- * *B. They increase stopping time, reducing impact force* ☒
- * C. They cushion visually
- * D. They keep you awake

25. When you catch a fast raw egg without breaking it, you pull your hand backward while catching. This works because:

- * A. Gravity reverses
- * B. It increases the final velocity
- * *C. It extends the contact time, reducing the average force* ☒
- * D. It reduces the egg's mass

26. Two cars of equal mass and speed collide head-on and stop. Compared with one car hitting a wall at the same speed, the force on each driver is:

- * *A. Essentially identical* ☒
- * B. Half
- * C. Double
- * D. Zero

27. Which correctly ranks the stopping force on an identical passenger from LOWEST to HIGHEST?

- * A. Dashboard → Airbag → Seatbelt
- * B. All equal
- * *C. Airbag + Seatbelt → Seatbelt only → Dashboard/Windshield* ☒
- * D. Wall → Airbag + Seatbelt → Seatbelt only

28. In a student investigation with marbles of different masses rolled at different speeds, the pattern that will emerge is:

- * A. Only mass matters
- * *B. Higher mass $\times$ higher speed produces greater momentum and greater impact* ☒
- * C. Only speed affects impact
- * D. No pattern

29. A firefighter turns on a powerful hose and feels it push backward strongly. This is a direct result of:

- * A. Water being heavy
- * B. Friction
- * C. Air pressure
- * *D. Conservation of momentum* ☒

30. The key difference between elastic and inelastic collisions is:

- * A. Momentum is conserved only in elastic collisions
- * *B. Total kinetic energy is conserved only in elastic collisions; momentum is conserved in both* ☒
- * C. Inelastic collisions have no momentum

* D. There are no forces in elastic collisions

Group 3· LARGE-SCALE GENERATION & DISTRIBUTION OF ELECTRICITY (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

The correct order of electricity from the power plant to your home is:

A. Meter → plant → lines

B. Power plant → step-up substation → transmission lines → step-down substation → distribution lines → home/meter ☒

C. Home → substation → plant

D. Transmission → plant → home

A device that changes AC voltage up or down very efficiently is a:

A. Transformer ☒

B. Generator

C. Motor

D. Resistor

Electricity is transmitted across the country at very high voltage mainly to:

A. Make electricity travel faster

B. Kill birds

C. Reduce energy lost as heat in transmission wires ☒

D. Make it brighter

Before entering houses, voltage is stepped down to about:

A. 12 V

B. 10,000 V

C. 1 V

D. 220–240 V ☒

A generator converts:

A. Mechanical motion into electrical energy ☒

B. Electricity into motion

C. Heat into light

D. Chemical energy into light

An electric motor converts:

A. Motion into electricity

B. Heat into sound

C. Electrical energy into mechanical motion ☒

D. Light into electricity

Which safety device melts and breaks the circuit if current becomes dangerously high?

A. Plug

B. Fuse ☒

C. Wire

D. Switch

Which reusable safety device automatically trips when too much current flows?

A. Fuse

B. Bulb

C. Outlet

D. Circuit breaker ☒

The third round pin of a three-prong plug is connected to:

A. Ground (Earth) wire ☒

B. Neutral wire

C. Live wire

D. Fuse

The device that measures your household's electricity consumption is the:

A. Generator

B. Transformer

C. Kilowatt-hour (kWh) electric meter ☒

D. Substation

Step-up transformers at power plants:

A. Lower voltage for homes

B. Generate electricity

C. Raise voltage for long-distance transmission ☒

D. Stop current

Step-down transformers at substations:

A. Lower transmission voltage to safer distribution levels ☒

B. Raise voltage

C. Store electricity

D. Burn waste

One major similarity between a motor and a generator is that:

A. Both plug directly into a wall outlet

B. Both have coils, magnets, and rotating parts ☒

C. Both always use batteries

D. Both produce electricity

One major difference between a motor and a generator is:

A. A generator has no magnets

B. There is no difference

C. A motor has no coils

D. A motor uses electricity to produce motion, while a generator uses motion to produce electricity ☒

Octopus wiring is dangerous because:

A. Too much current can overheat wires and start a fire ☒

B. It looks untidy

C. It uses more voltage

D. It breaks plugs

Why should you never use water to put out an electrical fire?

A. It makes the fire hotter

B. Water conducts electricity and can cause electrocution ☒

C. It damages the floor

D. It is expensive

What is the safest action if someone is being electrocuted while touching a live wire?

A. Turn off the main breaker first, then separate them using dry wood ☒

B. Grab them with your bare hands

C. Pour water on the wire

D. Pull the wire with metal

Why are transmission lines made with thick aluminum cables on tall towers?

A. Birds like them

B. Steel is cheaper

C. Aluminum is a good conductor, and tall towers provide safe clearance ☒

D. They look better

A GFCI/ELCB outlet protects people by:

A. Increasing power

B. Making water safe

C. Lowering voltage

D. Detecting current leakage and shutting off the circuit quickly ☒

The simplest way to reduce household electricity use is to:

A. Leave lights on

B. Use LED bulbs, switch off unused appliances, and use inverter appliances ☒

C. Open the refrigerator often

D. Use more extension cords

Why does high voltage reduce transmission losses?

A. Higher voltage means lower current, reducing I^2R power loss ☒

B. High voltage cools wires

C. Voltage removes resistance

D. Resistance disappears

Which correctly describes the roles of NGCP and distribution utilities?

A. Distribution utilities generate electricity

B. Both perform the same job

C. NGCP handles high-voltage transmission, while distribution utilities deliver electricity to consumers

☒

D. NGCP bills homes directly

Inside a power plant generator, you would find:

A. Solar panels

B. Batteries charging

C. A turbine spinning an electromagnet inside coils to produce AC electricity ☒

D. Chemical reactions

Which statement about motors and generators is TRUE?

A. They share no parts

B. A generator requires electricity as input

C. A motor works without energy input

D. A DC motor can act as a generator when mechanically spun ☒

Why does the Philippines use alternating current (AC) for the national grid?

- A. AC can be easily stepped up and down using transformers ☒
- B. Edison preferred AC
- C. AC is always safer to touch
- D. DC cannot run motors

A properly wired Philippine home has Live, Neutral, and Ground wires. Which statement is correct?

- A. Neutral is always dangerous
- B. Live carries 220 V, Neutral is the return path, and Ground is for safety ☒
- C. All three wires are the same
- D. Ground carries normal current

Why are circuit breakers preferred over fuses in modern homes?

- A. They are cheaper
- B. They look better
- C. They use less metal
- D. They can be reset after tripping and provide reliable protection ☒

Reducing household electricity use provides which benefit?

- A. Lower electricity bills and reduced environmental impact ☒
- B. Only lower bills
- C. Only global environmental benefits
- D. No benefits

Why are substations built at different voltage levels?

- A. To waste money
- B. To store electricity
- C. To match the safest and most efficient voltage for each stage of distribution ☒
- D. To confuse consumers

Which activity best models the journey of electricity from the power plant to homes?

- A. Testing a fuse
- B. Lighting a bulb with a battery
- C. Spinning a motor
- D. Small generator → step-up transformer → transmission wire → step-down transformer → bulb and meter ☒

Group 4• RENEWABLE AND NON-RENEWABLE ENERGY (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

Group 4 – Renewable and Non-Renewable Energy

Energy sources that take millions of years to form and are gone once used are called:

- A. Infinite
- B. Non-renewable (fossil fuels) ☒
- C. Free
- D. Renewable

Energy sources replenished naturally within a human lifetime are:

- A. Renewable ☒
- B. Fossil fuels
- C. Exhaustible
- D. Non-renewable

The top source of electricity in the Philippines today is:

- A. Solar
- B. Wind
- C. Coal ☒
- D. Tidal

The Philippines ranks second worldwide in which renewable energy source?

- A. Oil
- B. Coal
- C. Solar
- D. Geothermal ☒

Burning fossil fuels releases large amounts of which greenhouse gas?

- A. Carbon dioxide (CO₂) ☒
- B. Oxygen
- C. Nitrogen
- D. Hydrogen

Energy from falling or flowing water in dams like Angat and Magat is called:

- A. Diesel
- B. Coal
- C. Hydroelectric power ☒
- D. Nuclear

Energy captured directly from sunlight using solar panels is:

- A. Wind
- B. Solar ☒
- C. Biomass
- D. Geothermal

Energy from rice husks, bagasse, and crop waste is:

- A. Coal
- B. Natural gas
- C. Nuclear
- D. Biomass (Bioenergy) ☒

Which is a major non-renewable energy source in the Philippines?

- A. Natural gas ☒
- B. River flow
- C. Sunlight
- D. Wind

Renewable energy helps fight climate change because it usually:

- A. Releases more CO₂
- B. Adds methane
- C. Produces little to zero CO₂ during operation ☒
- D. Only makes oxygen

Which list contains only non-renewable energy sources?

- A. Solar, wind, hydro
- B. Coal, oil, natural gas, nuclear ☒
- C. Geothermal and biomass
- D. All of these

Which list contains only renewable energy sources used in the Philippines?

- A. Geothermal, hydro, solar, wind, biomass ☒
- B. Nuclear and fusion
- C. Hydrogen only
- D. Coal, oil, gas

The biggest harmful effect of coal-fired power plants is:

- A. Noise
- B. Air pollution, high CO₂ emissions, and toxic ash ☒
- C. Water use only
- D. Appearance only

A major disadvantage of large hydroelectric dams is:

- A. They produce more CO₂ than coal
- B. They block the wind
- C. They always cause earthquakes
- D. They can flood communities and affect rivers and ecosystems ☒

Why is geothermal energy ideal for the Philippines?

- A. Its location on the Pacific Ring of Fire provides abundant underground heat ☒
- B. It uses imported fuel
- C. It prevents typhoons
- D. The country has no volcanoes

A key economic benefit of increasing renewable energy is:

- A. It creates no jobs
- B. It reduces fuel imports and protects against fuel price increases ☒
- C. It always increases costs
- D. It raises import bills

Solar and wind energy are called intermittent because:

- A. They never work
- B. They only work at night
- C. Their output depends on sunlight and wind conditions ☒
- D. They break often

Natural gas is considered a transition fuel because:

- A. It lasts forever
- B. It is renewable
- C. It produces zero pollution
- D. It emits less CO₂ than coal but is still a fossil fuel ☒

Replacing coal with renewable energy improves public health by:

- A. Reducing air pollution that causes lung and heart diseases ☒
- B. Increasing mining
- C. Making no difference
- D. Raising electricity bills

Wind farms like those in Ilocos Norte have which advantage?

- A. They work all the time
- B. They use free wind and produce very low emissions ☒

C. They use large amounts of water

D. They block sunlight

Which comparison best describes renewable and non-renewable energy?

A. Both run out quickly

B. Their impacts are identical

C. Renewables cost more to build but are cleaner and cheaper to operate long-term ☒

D. Renewables always cost more forever

Why is the Philippines increasing its renewable energy targets?

A. To stop all power generation

B. To favor only one technology

C. To improve energy security, reduce emissions, and lower fuel imports ☒

D. Because the targets are random

Geothermal and hydroelectric power are called baseload renewables because they:

A. Can provide reliable electricity continuously ☒

B. Only work at noon

C. Need imported fuel

D. Cost more than coal

A major fairness issue in building large energy projects is:

A. Everyone benefits equally

B. There are never complaints

C. Prices are always lower

D. Local and indigenous communities may experience most of the negative impacts ☒

The best way to evaluate an energy source is by considering its:

A. Total cost, including environmental and health impacts ☒

B. Name

C. Advertisements

D. Appearance

Coal ash and mining waste are dangerous because they:

A. Help crops grow

B. Can release toxic substances into soil and water ☒

C. Break down quickly

D. Are completely harmless

How does the choice of energy source affect electricity delivered to homes?

A. Only the meter changes

B. Only wire color changes

C. It affects cost, reliability, emissions, and electricity bills ☒

D. It has no effect

Which energy mix is most suitable for reliable and clean electricity in the Philippines?

A. Wind only

B. Biomass only

C. Solar only

D. Geothermal and hydro supported by solar, wind, storage, and grid upgrades ☒

Why are renewable energy prices generally more stable?

A. They do not require fuel purchases after installation ☒

B. Government permanently fixes the price

C. Oil companies control them

D. They are always subsidized

Why is it important for Grade 10 students to learn about energy sources?

A. It is only for examinations

B. Future citizens will make decisions that affect energy, the environment, and society ☒

C. Adults make all decisions

D. It has no practical use